

YOLO Model Testing Report

Model:	yolov8x
Model Size:	130.5 MB
Parameters:	68.2 M
FLOPs (640x640):	258.55 GFLOPs
Run Name:	yolov8x_testing_20251123_230505
W&B Run Name:	yolov8x_bdd100k_yolo_test_20251123_230505
Timestamp:	2025-11-23 23:19:46
Dataset:	bdd100k_yolo - test split
Images Processed:	20000
Total Execution Time	856.84s
IoU Threshold:	0.5

Inference Performance

Metric	Value
Average Inference Time	39.44 ms per image
FPS (Frames Per Second)	25.35

Note: Inference time includes preprocessing, model inference, and postprocessing.

Overall Accuracy Metrics

Metric	Value
Precision	0.3255
Recall	0.6054
F1-Score	0.4233
mAP@0.5	0.1019

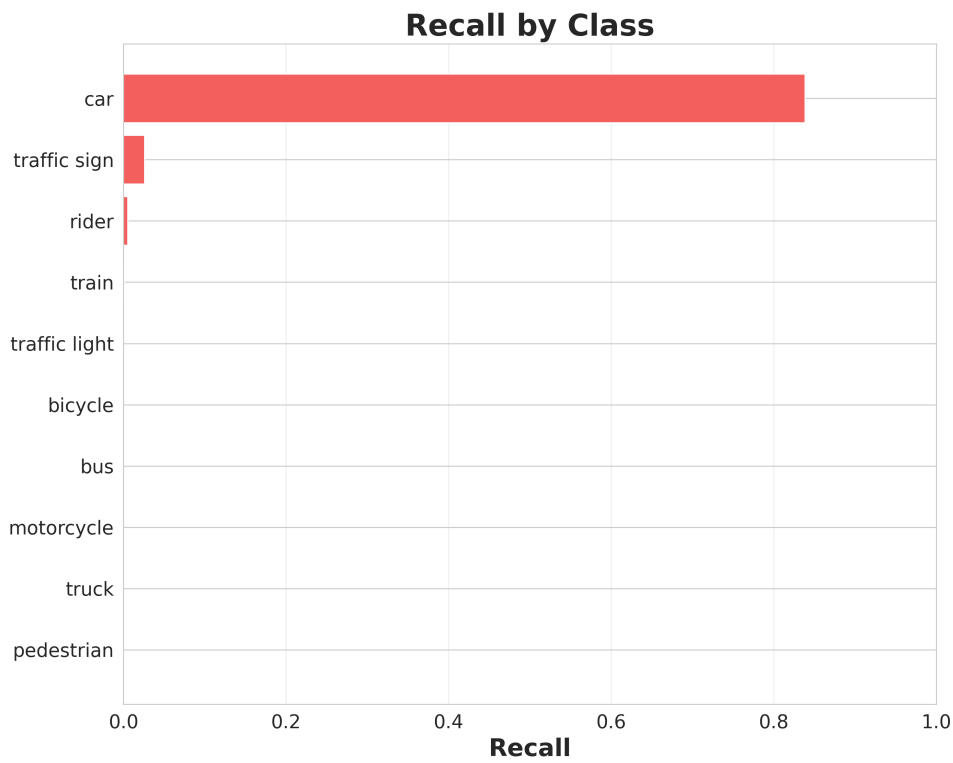
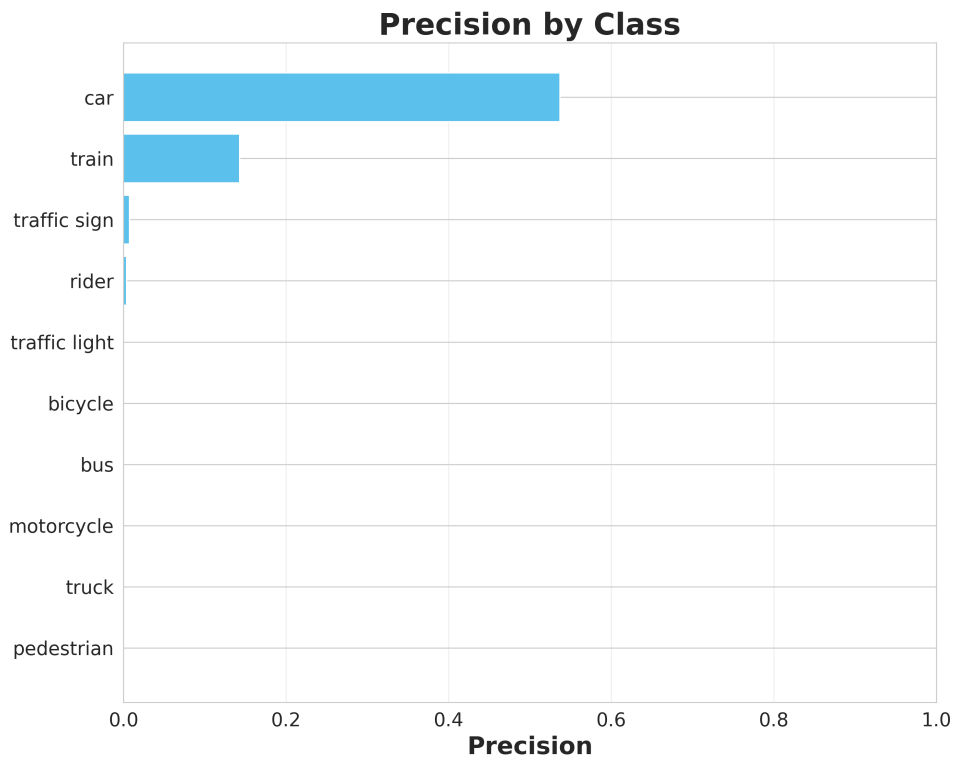
mAP@0.5:0.95	0.0608
True Positives	110704
False Positives	229437
False Negatives	72158

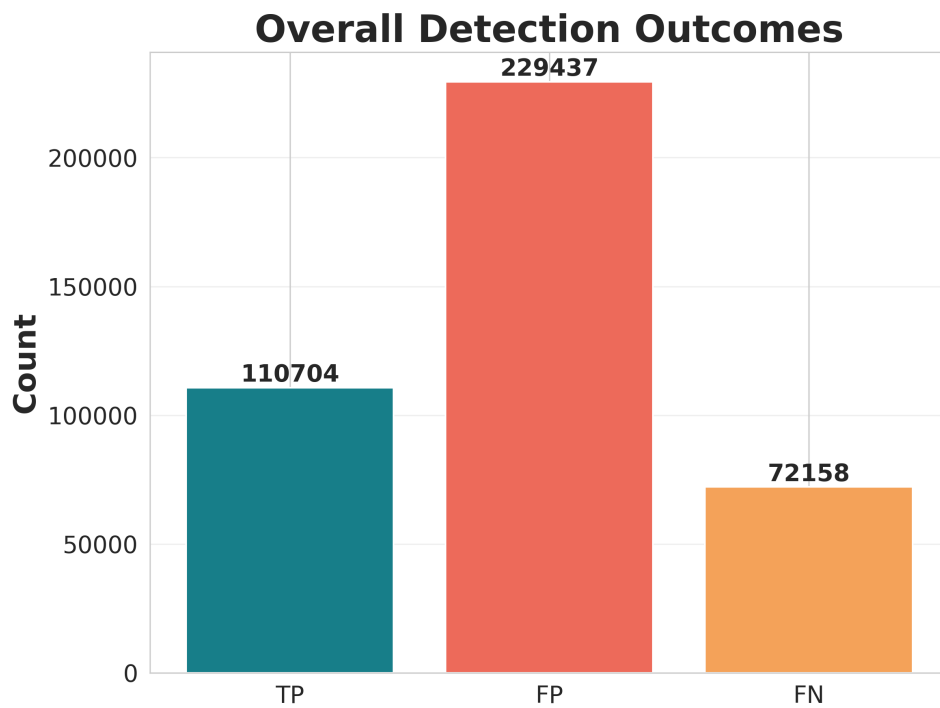
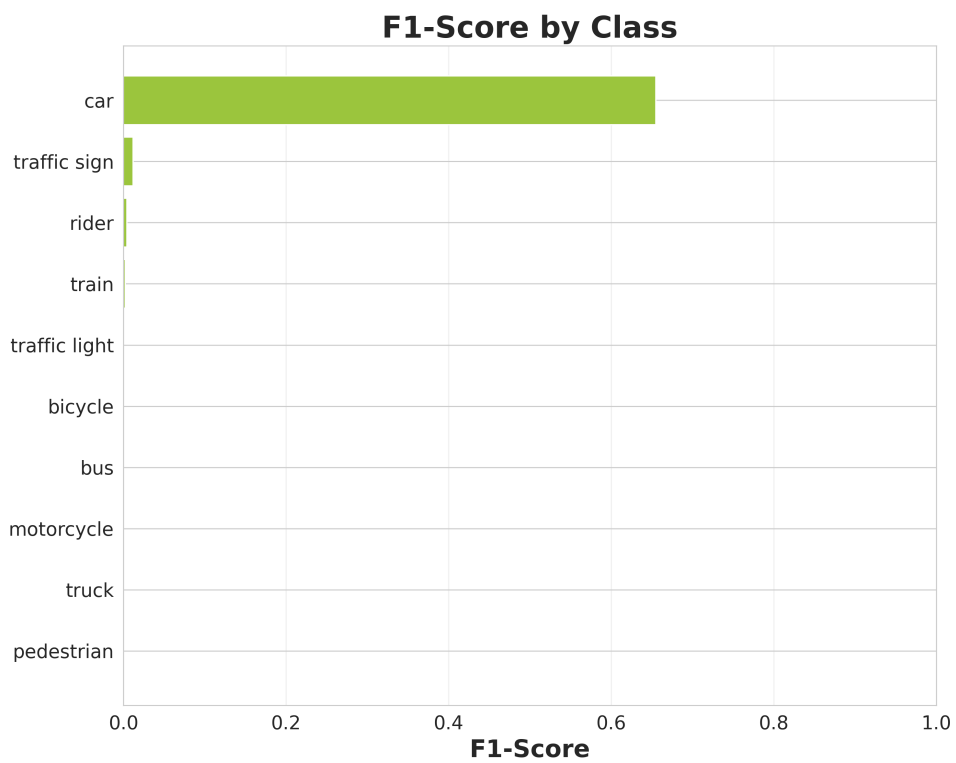
Confusion Matrix

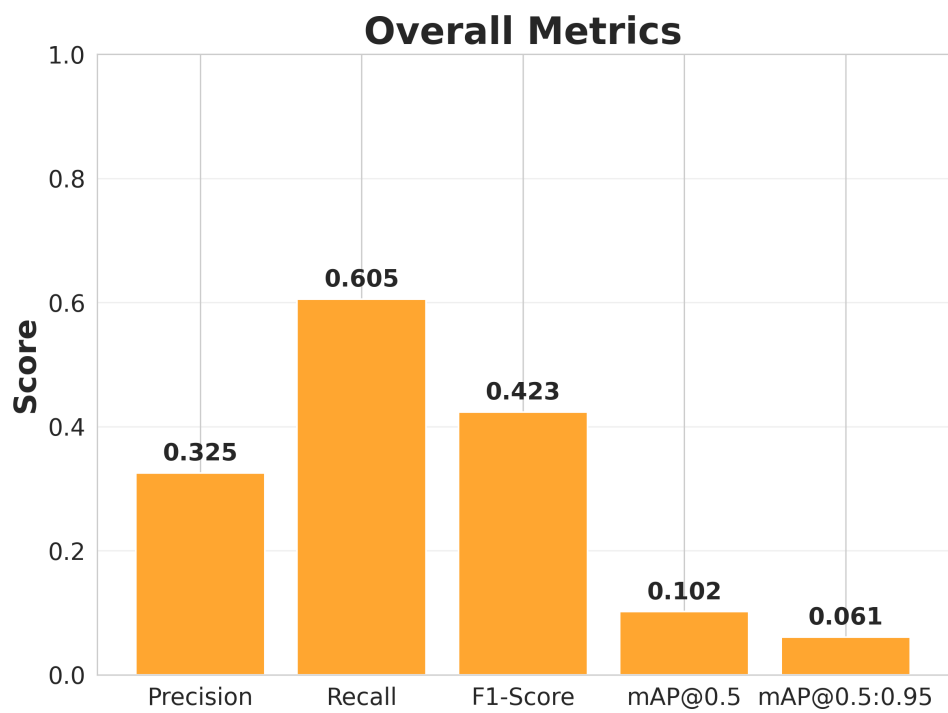
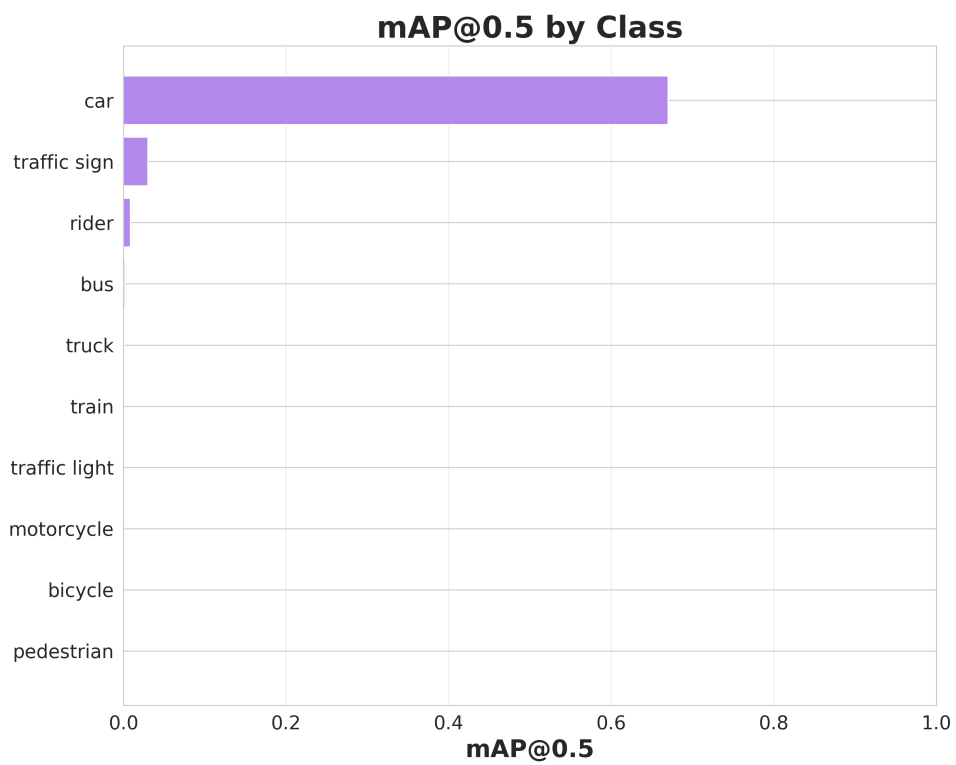
Confusion Matrix (yolov8x validation)										
True Class	pedestrian	692	92	1					5	18
	rider	5	1							
	car	15	110165	800	190				20	83
	truck	8	13							1
	bus									
	train		444	489	1511	4				4
	motorcycle		4	8	6	8				1
	bicycle	1	4063	4228	338	2			2	17
	traffic light									
	traffic sign		8	2		3			13805	530
		Predicted Class								
	pedestrian	rider	car	truck	bus	train	motorcycle	bicycle	traffic light	traffic sign

Correct Predictions (Diagonal Sum): 110704
Total Matched Predictions: 387601

Performance Visualizations







Per-Class Performance

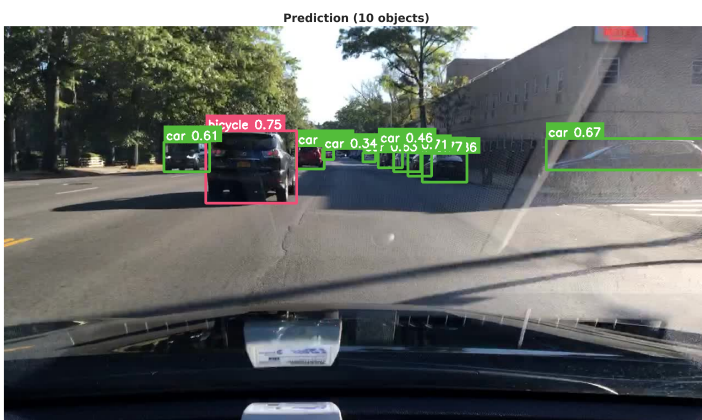
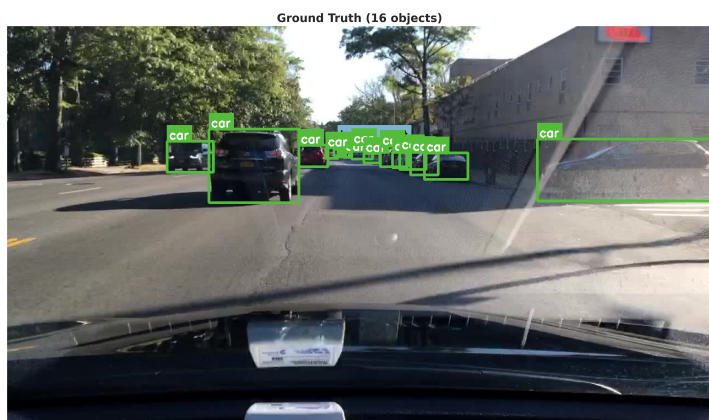
Class	TP	FP	FN	Precision	Recall	F1-Score	mAP@0.5
pedestrian	0	0	14880	0.0000	0.0000	0.0000	0.0000
rider	5	1288	903	0.0039	0.0055	0.0045	0.0087
car	110165	94928	21186	0.5371	0.8387	0.6549	0.6703
truck	0	8701	512	0.0000	0.0000	0.0000	0.0011
bus	0	3217	21	0.0000	0.0000	0.0000	0.0021
train	4	24	3129	0.1429	0.0013	0.0025	0.0005
motorcycle	0	0	155	0.0000	0.0000	0.0000	0.0000
bicycle	0	0	11725	0.0000	0.0000	0.0000	0.0000
traffic light	0	52817	17	0.0000	0.0000	0.0000	0.0000
traffic sign	530	68462	19630	0.0077	0.0263	0.0119	0.0304

Sample Predictions: Ground Truth vs Model

Sample #1: e4130321-bb89977e.jpg

Ground Truth Objects: 16 | Predicted Objects: 10

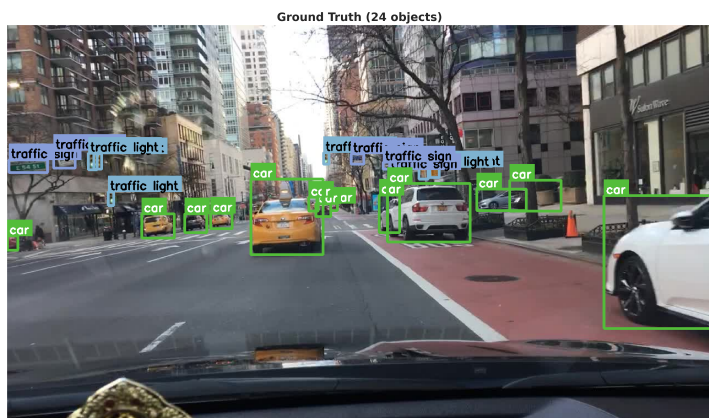
Weather: undefined | Scene: highway | Time of Day: daytime



Sample #2: cbdc7b52-739066ba.jpg

Ground Truth Objects: 24 | Predicted Objects: 13

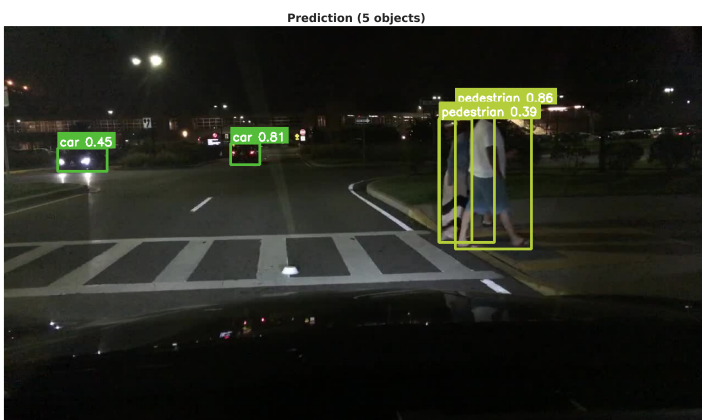
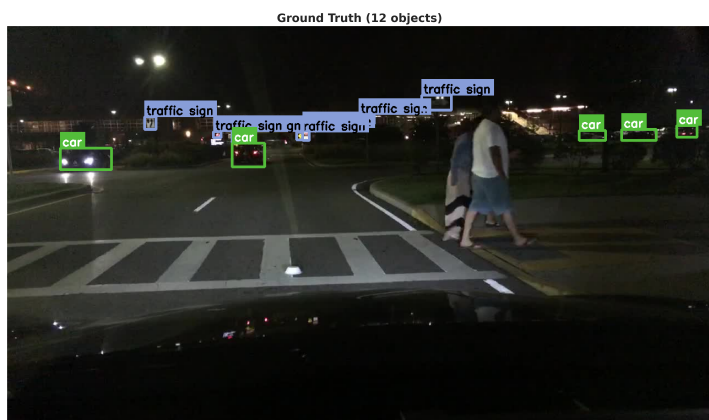
Weather: undefined | Scene: city street | Time of Day: daytime



Sample #3: d011f521-0314b67c.jpg

Ground Truth Objects: 12 | Predicted Objects: 5

Weather: clear | Scene: city street | Time of Day: night

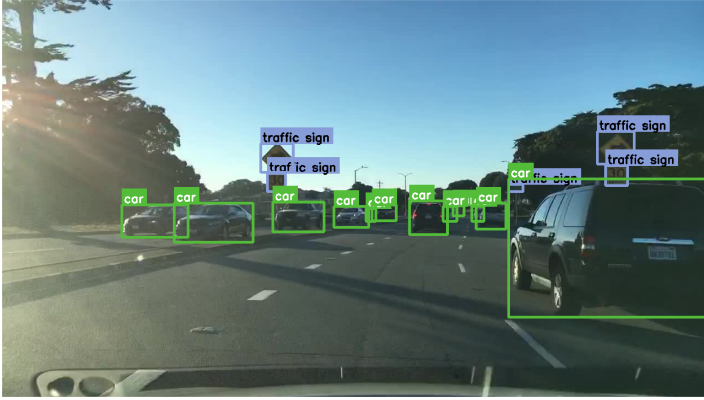


Sample #4: e1c2001c-e3f0a3c4.jpg

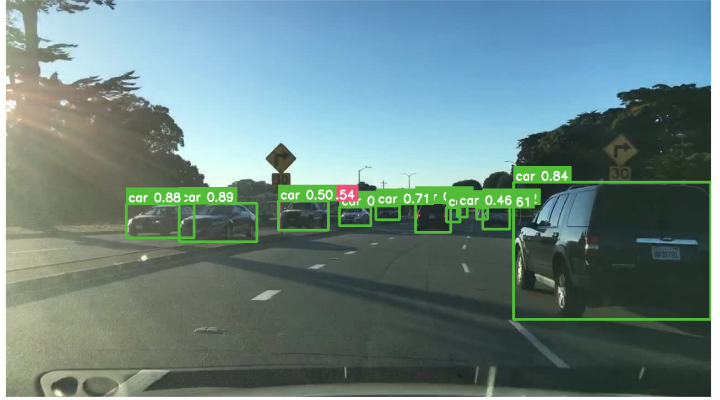
Ground Truth Objects: 17 | Predicted Objects: 12

Weather: clear | Scene: highway | Time of Day: daytime

Ground Truth (17 objects)



Prediction (12 objects)

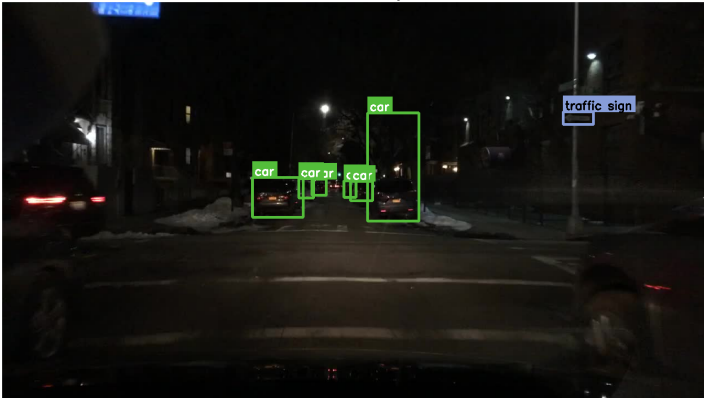


Sample #5: d06b1bf8-eafc4987.jpg

Ground Truth Objects: 7 | Predicted Objects: 7

Weather: snowy | Scene: residential | Time of Day: night

Ground Truth (7 objects)



Prediction (7 objects)

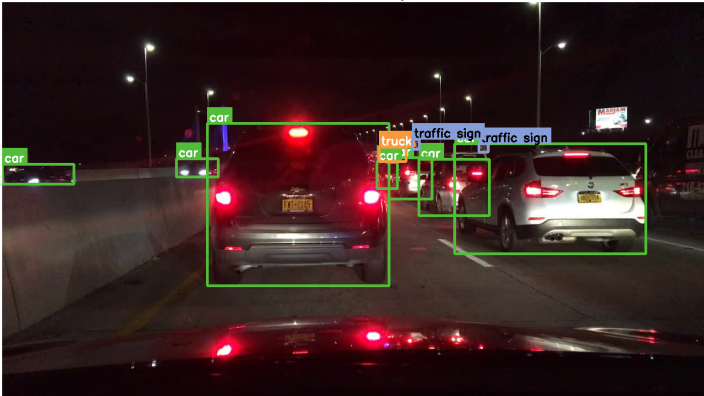


Sample #6: d1c9d617-da1557cd.jpg

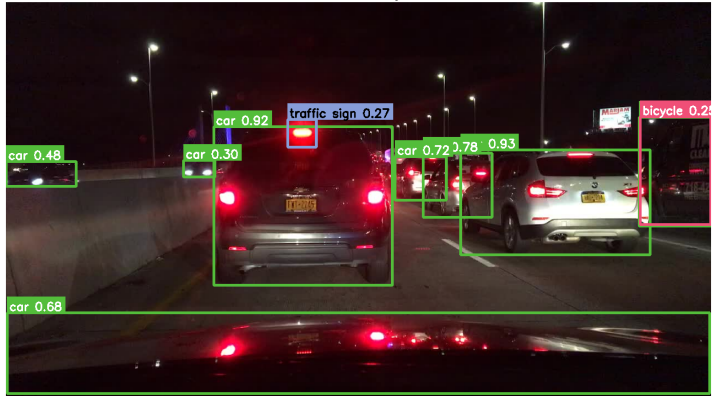
Ground Truth Objects: 10 | Predicted Objects: 10

Weather: clear | Scene: highway | Time of Day: night

Ground Truth (10 objects)



Prediction (10 objects)



Generated by YOLO Quick Test Script